==========

PlayList : spring boot full course

channel : india free internship

youtube channel https://www.youtube.com/@IndiaFreeInternship

source Code: spring boot source Code

==========

SESSION 1

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1. SPRING FRAMEWORK
2. SPRING BOOT
3. SRPING DATA JPA
4. SPRING WEB MVC
5. SPRING REST
6. SPRING SECURITY
7. MICROSERVICE
8. SPRING BATCH
9. SPRING SCHEDULAR

==========

SESSION 2

=========

Technolgies : webPage(servlets) +Database(JDBC/JPA) + Email (Java Mail) + Message(JMS)

Limitations:

- Start coding from line zero
- Development takes more time
- More chance of error
- Design pattern applied manually

Framework:

- Ready-made structure
- In-built technologies
- Design patterns already implemented

RAD => Rapid Application Development

- Fast coding
- Less Error

Spring Framework:

- used for Web Application or Distributed appliation
- It has own Apis
- It Follow Container System

Spring Container

-Find/Scan Classes[spring bean]
- Create Object
- provide DATA
-Link Object based on Relation
- Finally Destroy

Depedency Injection (DI)

Depedency Injection is a concept/theory which is implemented by spring IoC (Inversion of control) also called as spring container.

Theory	*programming*
oops	*core java*
ORM	*HIbernate with JPA*
DI	*Spring Ioc(Inversion of control)*

---

session 3

=========

Application

==========

3 Type of application

(i) Web Application

(ii) Distributed Application

(iii) Microservices Application

(i) Web Application : Collection of web pages run under webserver.

Application : - n services

Project :- n Module

eg : Gmail Application

User(Register and Login) , Inbox, Sent, Drfats, Setting, ... etc

Amazon Application

User(Register and Login) , search, Cart, Payment, ...etc

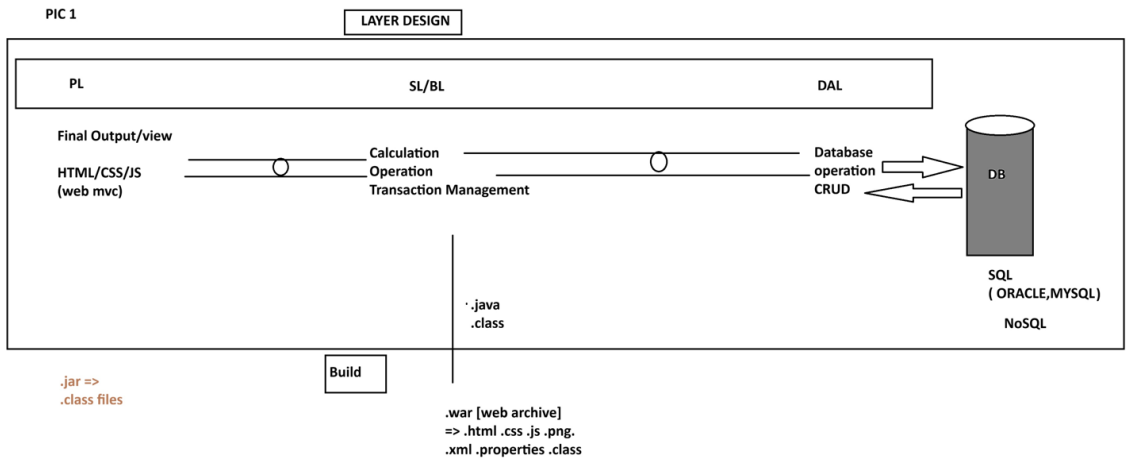
=> Implementation of service / module: - This is implemented using 3 Layers Design.

1. DAL : Data Access Layer (Database Operations)

2. SL : Service Layer (Calculation/Operations/txn mngmt)

3. PL : Presentation layer (view /Dynamic web pages/MVC)

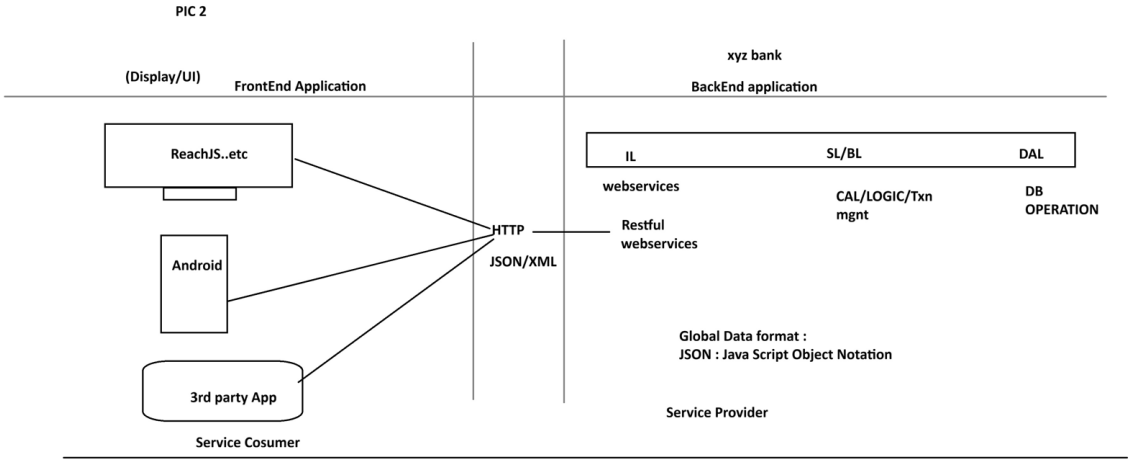
PIC 1



==========

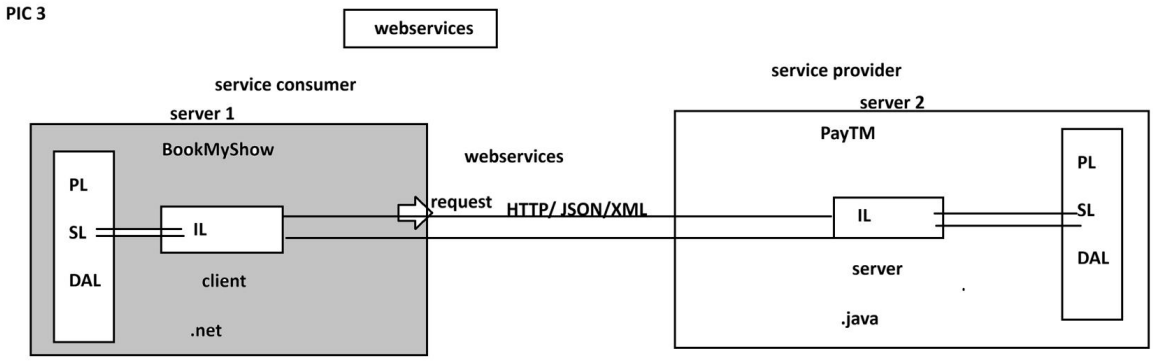
(ii) Distributed Application : An Application that runs at mutiple devices is called as distributed appliation.
xyz Bank application
This appliation can be accessed in different ways. Example.
eg:
1. Manually Process
2. NetBanking (Web App)
3. Mobile Banking
4. IVR
5. ATM / DC
6. 3rd party app (Gpay, PayPhone..etc)
.....

-But here business logics exist at backend application.
- Fontend applications are implemented using Angular, Android...etc
- App connected using HTTP Protocol using Global Data Format (JSON/XML)
PIC 2



WebServices:
==========
Intergration of multiple (at least two) applications which runs on different servers and implemented using different language (java, .net, php, python....etc)

PIC 3



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session 4

==========
==========
SPRING FRAMEWORK
==========
CORE
---
1. DI and IOC
2. Container Types
3. Annotation Configuration
4. @ComponentScan
5. @Component
6. @Value
7. Java Configuration
8. @Configuration
9. @Bean
10. @PropertySource
11. Environment
12. Lifecycle methods
13. Runners
14. @ConfigurationProperties
15. Properties file
16. YAML File
17. Profiles
18. Lombok API

==========
spring core chapter 1
==========
Core:
This module/chapter gives all rules and guidelines to work with spring container.
1. DI and IOC
==========

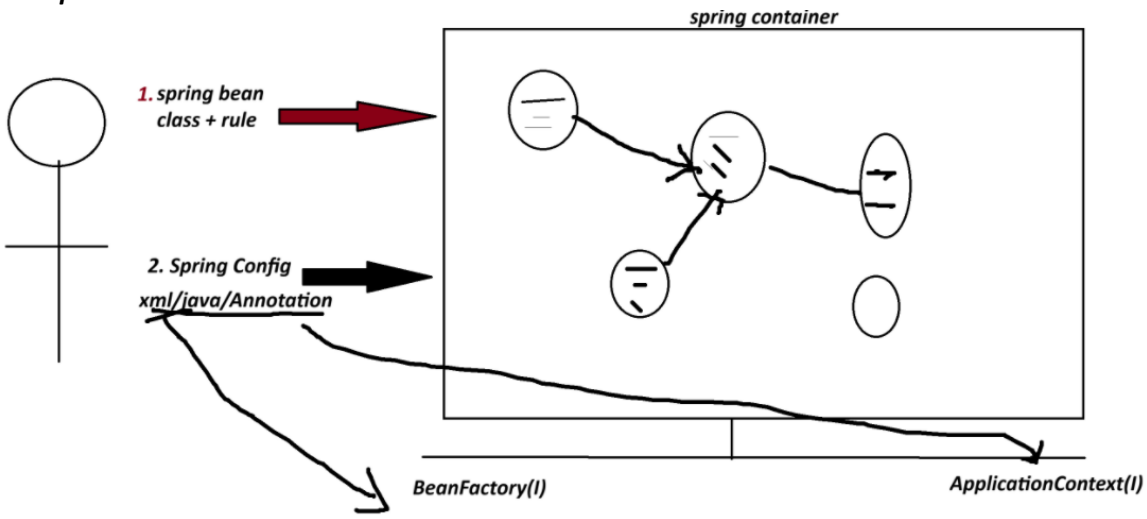
Spring Container :- It takes care of object.
1. Find/Scan our Classes(spring bean)
2. Create Object
3. Provide Data
4. Link Object
5. Destroy the Object

=>For this programmer has to provide two inputs-
1. Spring Bean : class+rules given by spring container.
2. Spring Configuration : This is main input cotains details like object name, relation with objects, data....etc.
xml/java/Annotation

*** Spring Container 2 Types:*
1. BeanFactory (Old container) : support only XML coonfiguration file.

2. ApplicationContext (New container) : support xml/java/Annotation configuration.

pic



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Dependeny Injection (DI)

Theory	*programming*
OOPs	*java*
ORM	*HIbernate with JPA*
DI	*Spring IoC(Spring Container) -> Inversion of Control*

=> Depedency :- An instance variable (Field) exist inside a class (spring bean)

1. Primitive Type Dependeny (PTD) :- If a variable is created using (8+1) datatype.
(byte, short, int, long, float, double, boolean, char +string)
(Their wrapper classess too)

2. Collection Type Depedency (CTD) :- If a variable created using(4) types:
(java.util.package)
(List,Map,Set and Properties)

3. Reference Type Depedency (RTD) :- If a variable is creatd using other class/interface

Injection(4)

======

=> It give/provide/assign existed provide data to Depedency(Inside object)

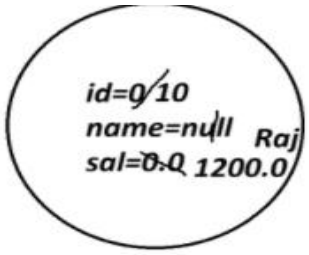
1. Setter Injection (SI) :- provide data to variable using its set method(setter).

2. Constructor Injection(CI): provide data creating object using parameterized constructor.

3. LookUpMethod Injection(LMI) :- Spring bean Scope
[When parent bean is sigleton and child bean is prototype)

4. Interface Injection(II) : - Not Exist in Spring F/W.

```
class Employee{
   int id;
   String name;
   double sal;
}
```




```
Employee e=new Employee()
```

```
class Product{
   int id; //PTD
   String name; //PTD
   Boolean exist; //PTD

   List<String> models; //CTD
   Map<String,Integer> data; //CTD

   Vendor vob; //RTD
   Service sob; //RTD
}
```

```
class Vendor{}

interface Service{}
```

SESSION 5

==========

1. Spring Bean

2. STS Installation

Spring Bean

=========

-It is class that follow rules given by Spring Container.

Rule

-----

a. class must be public Type.

b. Recomended to keep in a package (basePackage Rule)

ex :

com.app.service

com.app.controller

c. Fields(Variables) are option, if required recomended to keep private.

private Integer empId;

private String empName;

d. If variable present in class,

=>define default constructor with setter/getter

=>(or/and) Parameterized Constructor.

ex:

```
public class Employee{
private Integer id;
private String name;
public Empoyee(){
}
 public void setId(Integer id){
this.id=id;
 }

public Integer getId(){
 return id;
 }
}
```

e. We can override Object(c) methods:

toString() equals() hashCode()

[because non-private, non-final,non-static]

f. Inheritance Rule:

Allowed ->Only Spring API is allowed (ie spring F/W classes and interface)

not allowed-> EJB, Servlet

g. Annotation Rule:

Only we can add Spring API Annotation and Inetegration API Annotations

(Hibernate with JPA, Restful webServices annotations);

ex:

@Component, @Service, @Repository.. etc.

STS Installation

==========

step 1:

go to official website

spring.io

link : - https://spring.io/tools#eclipse

step 2 download STS according your OS

step 3

Rigth click on ZIP file

click on "Extract Here"

step 4

Open extracted folder

Run : SpringToolsForEclipse

Final STEP STS installed successfully.

==========

session 6

=========

1. Spring Bean

Simple Java class that follow rules given by container.

2. Spring Configuration File(XML)

It gives object details (name, data,links with other objects ... etc).

3. Test class=> Just find, container created or not? object is creatd or not?

==========

Spring core XML Tags

==========

<bean> - Object creation

<property> - calling set method

<constructor-arg> - call parameterized constructor

<value> - To provide data to Primitive Type.

<list> <set> <map> <props> -- To provide data to Collection type.

<ref/> -- To Link two objects

----

1. To create Object

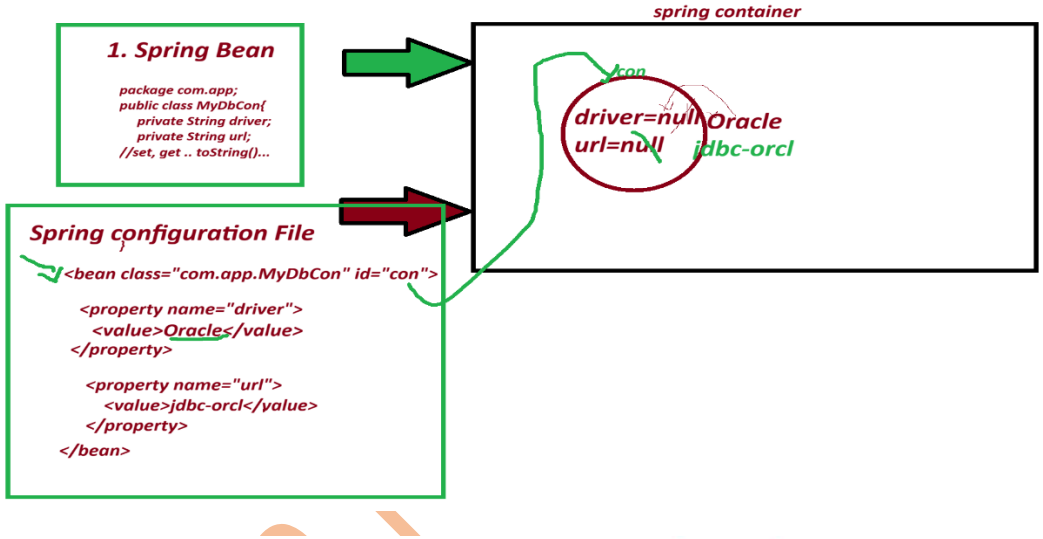
<bean id="objName" class="FullyQualifiedClassName"></bean>

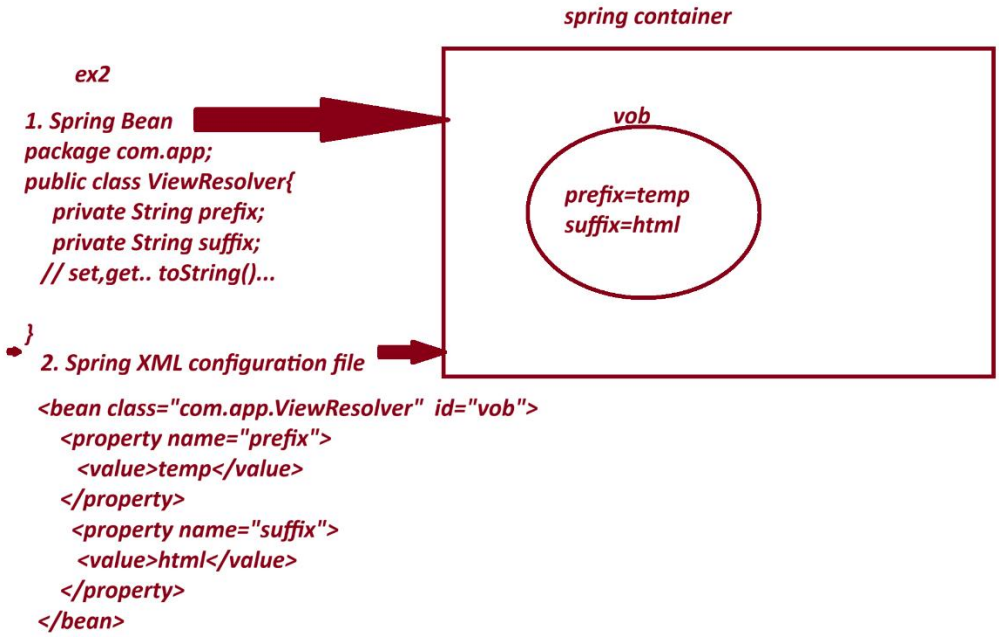
2. To call set method

<property name="variableName"></property>

3. To provide data to Primitive Variables

<value>data</value>





==========

1. Spring bean

```
package com.app;
public class PaymentService{
    private int token;
    private boolean active;
    private String provider;
   //set.. set toString..etc
}
```

2. Spring XML configuration file.

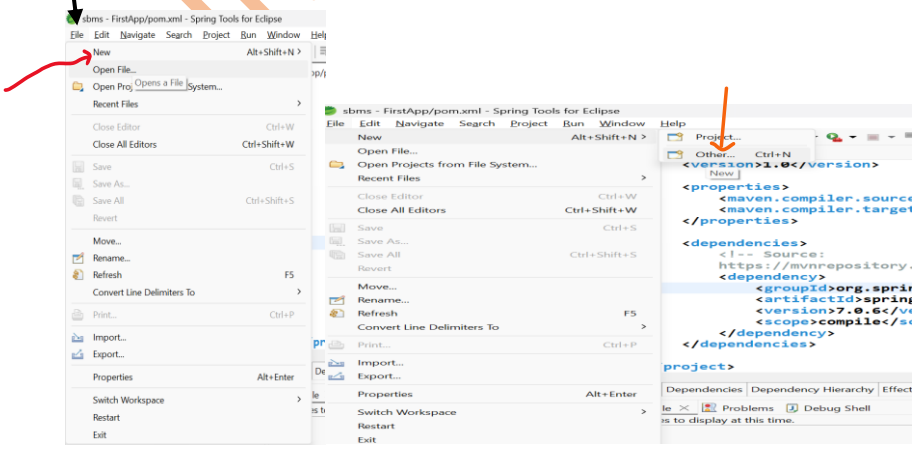
```
<bean class="com.app.PaymentService" id="psObj">
<property name="token">
<value>154678</value>
</property>
<property name="active">
<value>true</value>
</property>
<property name="provider">
<value>xyzBank</value>
</property>
</bean>
```

==========

CREATE MAVEN PROJECT WITH BELOW STEP :-

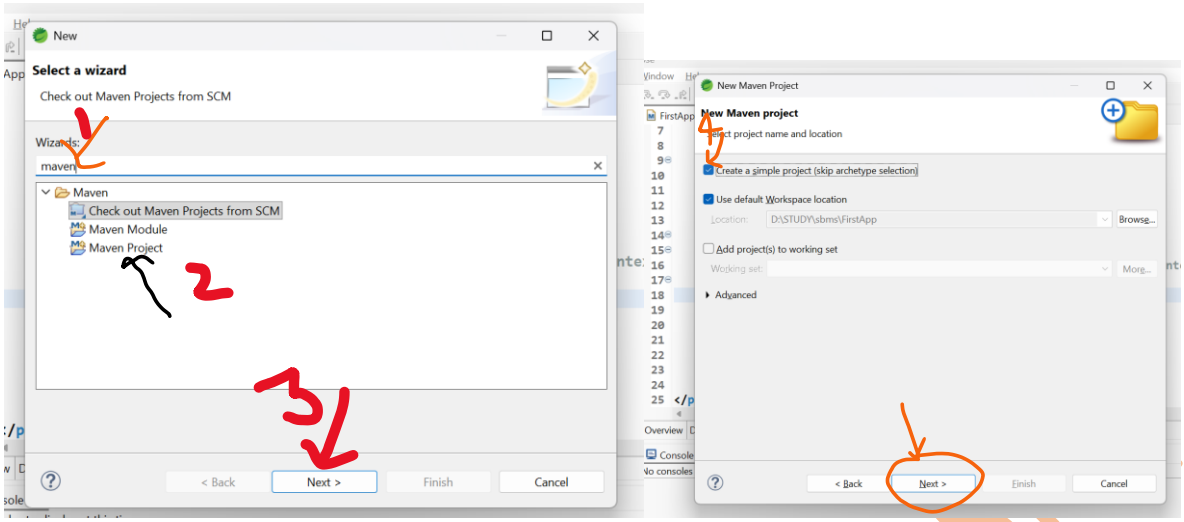
==========

1. Create Project



File-> New-> Other-> Type maven->choose maven project-> next->

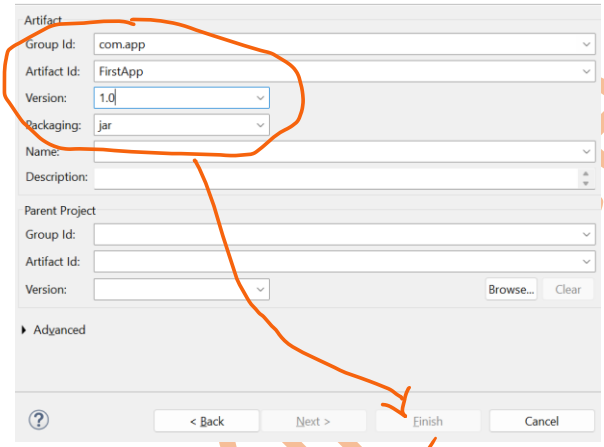
choose checkbox [v] create simple project -> next-> Enter details



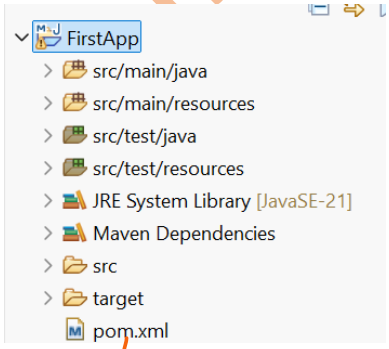
groupId : com.app

ArtficatId : FirstApp

version : 1.0



-> Finish



2. pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>FirstApp</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
        <!-- Source:
         https://mvnrepository.com/artifact/org.springframework/spring-context -->
  <dependency>
    <groupId>org.springframework</groupId>
     <artifactId>spring-context</artifactId>
     <version>7.0.6</version>
     <scope>compile</scope>
  </dependency>
</dependencies>
</project>
```

---

session 7

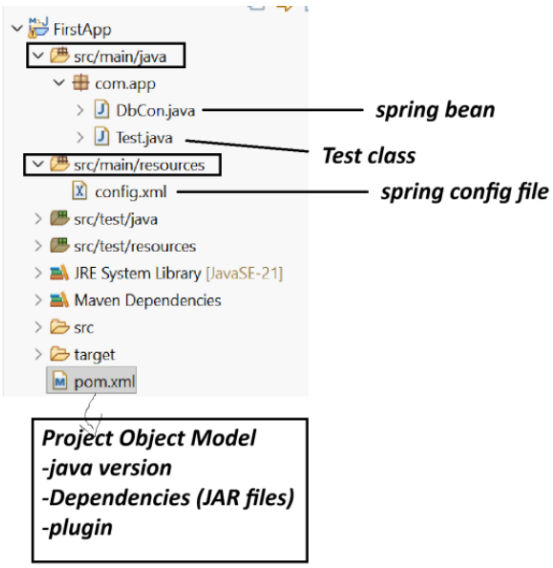
==========

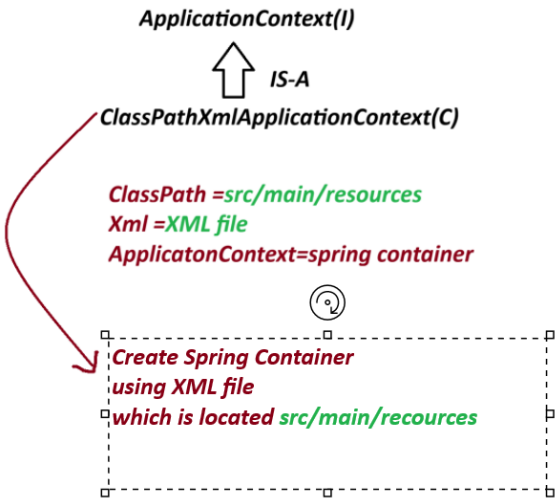
Spring Core First Application 3 Files.

1. Spring bean (.java)

2. Spring XML configuration (config.xml)

3. Test class(.java)





Note:-

1. Spring container are two Types.

(a) Legacy Container : BeanFactory(I) support only XML

(b) New Container : ApplicationContext(i) support all XML/java/Annotation config.

(c) ApplicationContext(I) : it is a interface.

we are using one of its Impl class:

ClassPathXmlApplicationContext(C).

==========

Create Maven Project with below Step:

==========

File>new>other>type maven>choose maven project>next>choose checkbox> create Simple project>next>Enter details

GroupId - com.app

ArtficatId - SecondApp

version - 1.0

----

code

SecondApp

==========

pom.xml

==========

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>SecondApp</artifactId>
<version>1.0</version>

<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
<!-- Source:https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

DbCon.java

```
package com.app;

public class DbCon {
  private String driver;
  private String url;
```

```
  //alt+shit+s
  public DbCon() {
super();
System.out.println("Constructor called");
  }
  public String getDriver() {
return driver;
  }
  public void setDriver(String driver) {
this.driver = driver;
System.out.println("Driver method called");
  }
  public String getUrl() {
return url;
  }
  public void setUrl(String url) {
this.url = url;
System.out.println("Url method called");
  }
  @Override
  public String toString() {
return "DbCon [driver=" + driver + ", url=" + url + "]";
  }

}
```

==========

config.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
   xsi:schemaLocation="
      http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">

   <bean class="com.app.DbCon" id="mobj">
      <property name="driver">
      <value>Oracle Driver</value>
      </property>

      <property name="url">
      <value>JDBC-ORCL</value>
      </property>
   </bean>

</beans>
======================

Test.java
---------------------
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {
//ctrl+shit+O
public static void main(String[] args) {
ApplicationContext ac=new ClassPathXmlApplicationContext("config.xml");
Object ob = ac.getBean("mobj");
System.out.println(ob);
}
}
```

---

output

Constructor called

Driver method called

Url method called

DbCon [driver=Oracle Driver, url=JDBC-ORCL]

---

session 8

==========

Spring Container (DI - Depedency Injection)

=>Depedency - Variables exist in class

3 type

---

1. Primitive Type (8+1) : byte, short, int, long, float, double, boolean, char, String

2. Collection Type (4) : Set, List, Map and Properties

3. Reference Type (x) : using a class/interface variable

=> Injection : provide Data

- Setter Injection

- constructor Injection

-

LookUpMethod Injetion

- Field Injection (Annotations)

==========

XML TAG

==========

<bean> - Object creation

<property> - calling set method

<constructor-arg> - call parameterized constructor

<value> - To provide data to Primitive Type.

<list> <set> <map> <props> -- To provide data to Collection type.

<ref/> -- To Link two objects

==========

Collection Configuration

==========

=> Spring container support working wiht collection configuration which are from java.util package

Collection Name	Collection Type	XML Tag Name
List	Interface	<list>
Set	Interface	<set>
Map	Interface	<map> and <entry>
Properties	Class	<props> and <prop>

List -> ArrayList

Set -> LinkedHasSet

Map -> LinkedHasMap

Properties --> NA

---

coding part collectionTypeS8

---

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>collectionTypeS8</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
```

```
</properties>

<dependencies><!-- Source:
https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

Spring bean

MyDetails.java

```
package com.app;

import java.util.List;
import java.util.Map;
import java.util.Properties;
import java.util.Set;

public class MyDetails {
//ctrl+shift+O
private List<String> data;
private Set<String> models;
private Map<Integer,String> design;
private Properties myinfo;

public MyDetails() {
super();
```

```
}

public List<String> getData() {
return data;
}

public void setData(List<String> data) {
this.data = data;
}

public Set<String> getModels() {
return models;
}

public void setModels(Set<String> models) {
this.models = models;
}

public Map<Integer, String> getDesign() {
return design;
}

public void setDesign(Map<Integer, String> design) {
this.design = design;
}
public Properties getMyinfo() {
return myinfo;
}public void setMyinfo(Properties myinfo) {
this.myinfo = myinfo;
}
@Override
public String toString() {
```

```
return "MyDetails [data=" + data + ", models=" + models + ", design=" + design + ", myinfo=" + myinfo +
"]";

}

}
```

---

xml config file

config.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"xsi:schemaLocation="http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">


<bean class="com.app.MyDetails" id="myinfo">

<property name="data">

<list>

<value>A</value>

<value>B</value>

<value>C</value>

<value>A</value>

</list>

</property>

<property name="models">

<set>

<value>M1</value>

<value>M2</value>

<value>M3</value>

<value>M1</value>

</set>

</property>
```

```
<property name="design">
<map>
<entry>
<key>
<value>101</value>
</key>
<value>AB</value>
</entry>
<entry>
<key>
<value>102</value>
</key>
<value>Suraj</value>
</entry>
<entry>
<key>
<value>103</value>
</key>

<null></null>
</entry>

</map>
</property>

<property name="myinfo">
<props>
<prop key="A1">B1</prop>
<prop key="A2">B2</prop>
<prop key="A3">B3</prop>
<prop key="A5"></prop>
</props>
```

```
</property>
</bean>
</beans>
```

Test File

Test.java

```
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {

public static void main(String[] args) {

ApplicationContext ac=new ClassPathXmlApplicationContext("config.xml");
//Object ob = ac.getBean("myinfo");
MyDetails ob = ac.getBean("myinfo",MyDetails.class);
System.out.println(ob);
System.out.println(ob.getData().getClass().getName());
System.out.println(ob.getModels().getClass().getName());
System.out.println(ob.getDesign().getClass().getName());
System.out.println(ob.getMyinfo().getClass().getName());
}

}
```

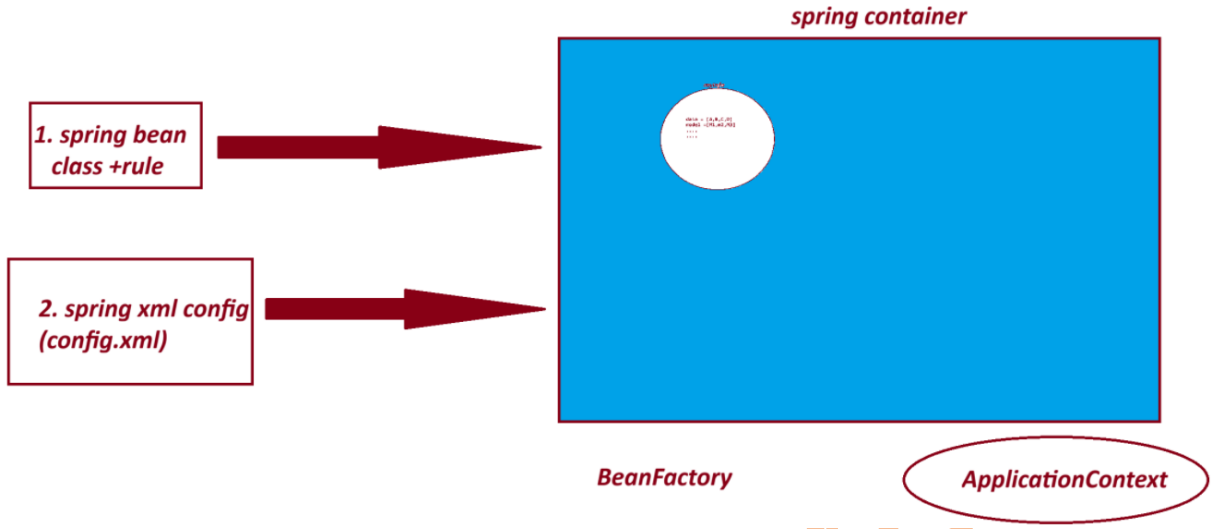
---

console output

```
MyDetails [data=[A, B, C, A], models=[M1, M2, M3], design={101=AB, 102=Suraj, 103=null},
myinfo={A1=B1, A2=B2, A3=B3, A5=}]
java.util.ArrayList
java.util.LinkedHashSet
```

java.util.LinkedHashMap

java.util.Properties



===

session 9

========

3. Reference Type Depedency - Setter Injection.

  ============================

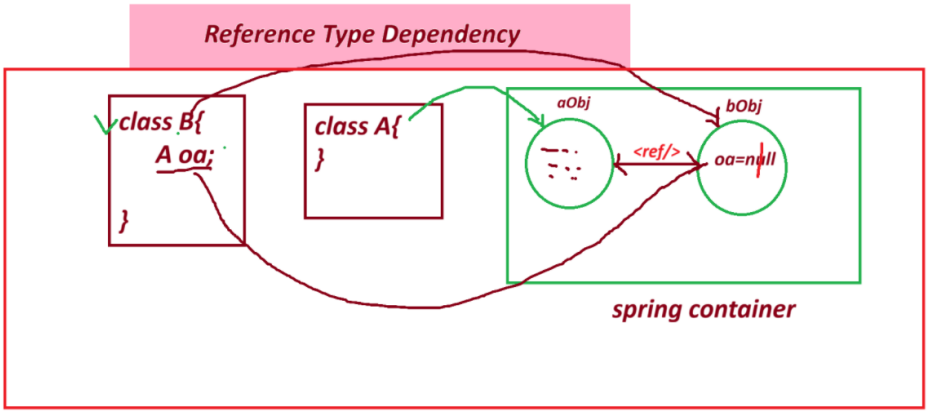
  => <ref/> : - tag is used to Link objects in case of HAS-A Relation.

        Syntax:

<property name="variableName">

<ref bean="childObjName"/>

</property/>



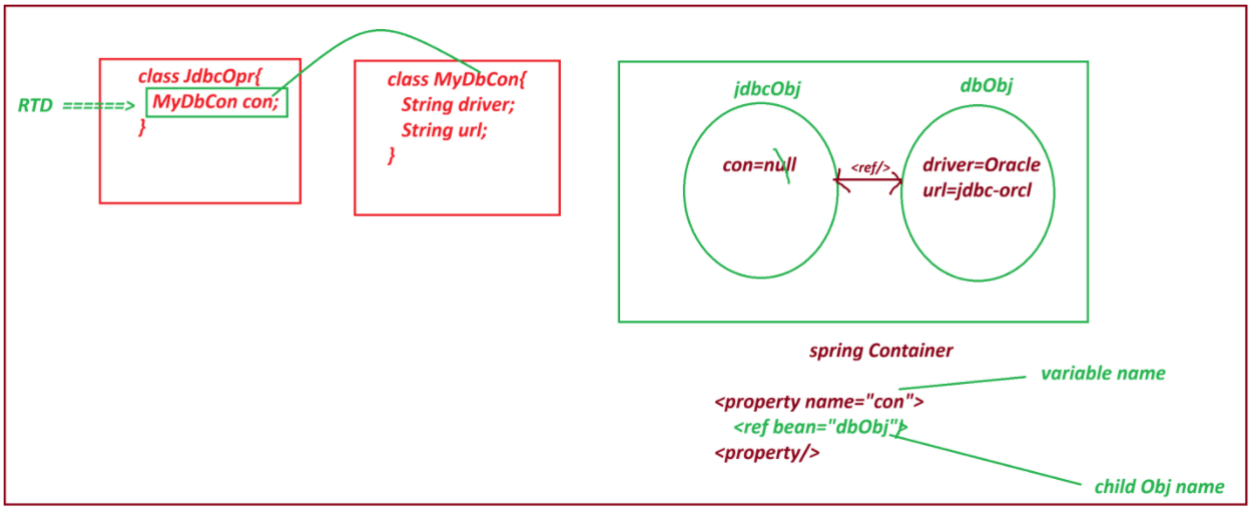
======================

- If we do not use <ref/> then ref variable holds default value null.

=> spring container creates objet to both classes and link them by using <ref/> tag.

=> if child object nam (ref bean name) is not valid or not exist then spring container throws exception like NoSuchBeanDefinitionException:

===



project : ReferenceTypeS9

----

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>ReferenceTypeS9</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>
<dependencies>
```

```
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>

</project>
```

JdbcOpr.java

```
package com.app;

public class JdbcOpr {
  private MyDbCon con; //RTD

  public JdbcOpr() {
super();
// TODO Auto-generated constructor stub
  }

  public MyDbCon getCon() {
return con;
  }

  public void setCon(MyDbCon con) {
this.con = con;
  }
```

```
  @Override
  public String toString() {
return "JdbcOpr [con=" + con + "]";
  }

}
```

==========

MyDbCon.java

```
package com.app;

public class MyDbCon {
  private String driver;
  private String url;

  public MyDbCon() {
super();
// TODO Auto-generated constructor stub
  }

  public String getDriver() {
return driver;
  }

  public void setDriver(String driver) {
this.driver = driver;
  }

  public String getUrl() {
return url;
  }
```

```
 public void setUrl(String url) {
this.url = url;
 }

 @Override
 public String toString() {
return "MyDbCon [driver=" + driver + ", url=" + url + "]";
 }

 //alt+shift+s

}
```

config.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
   xsi:schemaLocation="
      http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">

 <bean class="com.app.MyDbCon"  id="dbObj" >
     <property name="driver">
     <value>OracleDriver</value>
     </property>
      <property name="url">
      <value>JDBC-ORCL</value>
      </property>
 </bean>
  <bean class="com.app.JdbcOpr"  id="jdbcObj">
    <property name="con">
    <ref bean="dbObj"/>
```

```
    </property>
  </bean>


</beans>
```

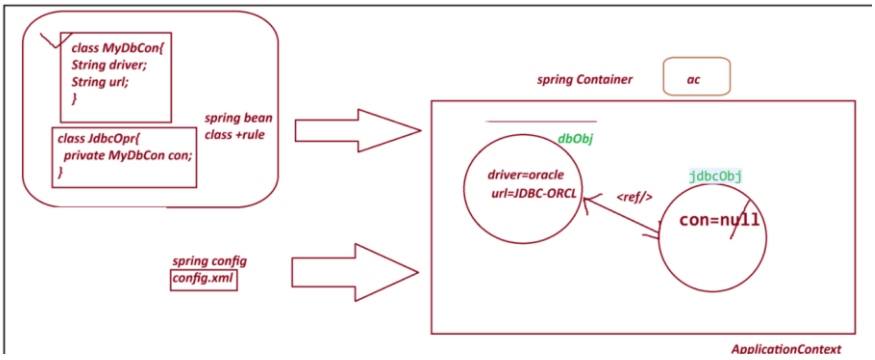
Test.java

```
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {
public static void main(String[] args) {
ApplicationContextac=new ClassPathXmlApplicationContext("config.xml");
JdbcOpr jdbc = ac.getBean("jdbcObj",JdbcOpr.class);
System.out.println(jdbc);
}
}
```

output

```
JdbcOpr [con=MyDbCon [driver=OracleDriver, url=JDBC-ORCL]]
```

==========



SI - Setter Injection :

Spring container creates object to given class using default constructor and provide data using setter method.

Employee emp=new Employee(); =======> <bean id="emp" class="Employee"/>

emp.setEmpId(10); =======> <property name="empId" value="10"/>

==========

***** Constructor Injection CI ************

==========

=> Sprint container creating object and providing data using Parameterized constructor.

Here , we use <constructor-arg> tag to indicate.

Employee emp=new Employee(10, "A"); =====>

```
<bean>
  <constructor-arg>
 <value>10</value>
</constructor-arg>
<constructor-arg>
 <value>A</value>
</constructor-arg>
</bean>
```

Note:

A. If a class has 3 param constructor then we need to pass <constructor-arg> 3 times in xml config file.

B. Data must be passed in order using <constructor-arg> as per constructor difined in class.

==========

code for CI (constructor Injection)

---

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>ConstructorInjectionS9</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>

</project>
```

Product.java

```
package com.app;

import java.util.List;

public class Product {

private String pcode;
```

```
private String model;
private List<String> data;
private Vendor vob;

public Product(String pcode, String model, List<String> data, Vendor vob) {
super();
this.pcode = pcode;
this.model = model;
this.data = data;
this.vob = vob;
}
@Override
public String toString() {
return "Product [pcode=" + pcode + ", model=" + model + ", data=" + data + ", vob=" + vob + "]";
}

}
```

Test.java

```
package com.app;

public class Vendor {

private String vcode;
private String vaddr;

public Vendor(String vcode, String vaddr) {
super();
this.vcode = vcode;
this.vaddr = vaddr;
}
```

```
@Override
public String toString() {
return "Vendor [vcode=" + vcode + ", vaddr=" + vaddr + "]";
}

}
```

---

config.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
   xsi:schemaLocation="
      http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">

 <bean class="com.app.Vendor" id="v1">
   <constructor-arg>
   <value>v1109</value>
   </constructor-arg>
   <constructor-arg>
   <value>IND</value>
   </constructor-arg>
 </bean>

 <bean class="com.app.Product" id="pobj">
  <constructor-arg>

<value>P01</value>
 </constructor-arg>


  <constructor-arg>
```

<value>HP-23444</value>

</constructor-arg>

<constructor-arg>

<list>

<value>AA</value>

<value>BB</value>

</list>

</constructor-arg>

<constructor-arg>

<ref bean="v1"/>

</constructor-arg>

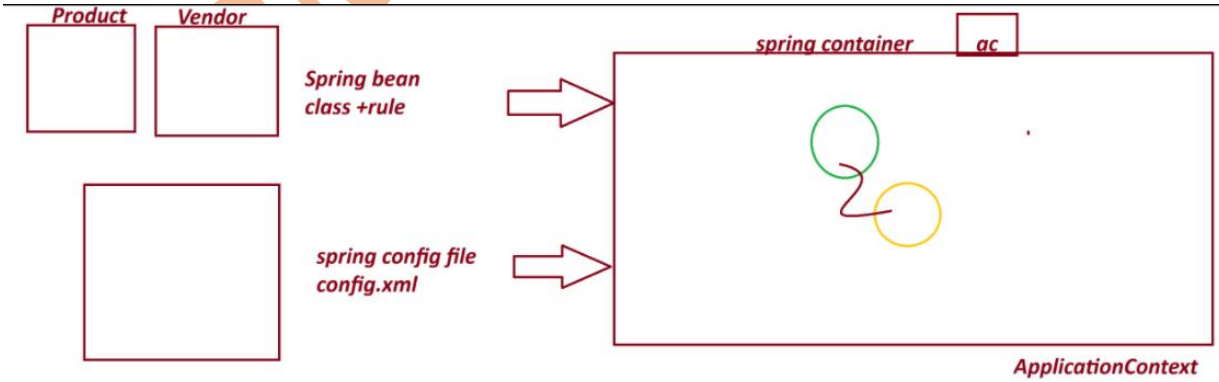
</bean>

</beans>

output

=========

Product [pcode=P01, model=HP-23444, data=[AA, BB], vob=Vendor [vcode=v1109, vaddr=IND]]



==========

session 10

==========

Spring Annotation Configuration

This is Easy to Configure

Faster in execution compare to xml.

==========

1. Stereotype Annotation:

These annotation informs Spring Container to create Object.

They are 5 Annotation given by Spring Framework.

@Component : Create object inside container.

@Repostitory : Creates object + Database operations

@Service : Creates object + Transaction/logic/calculation..etc

@Controller : Creates object + Http Protocol (Web App)

@RestController : Creates object + Http Protocol (webservices)

@ComponentScan

@ComponentScans

2. Data Annotation/ Basic Annotation

@Value Annotation is used to provide data (field inject) to variable.

It has 3 usages:

1. Hard coding direct value to a variable.

2. Reading Data from Properties/YAML Files

3. - SpEL (Spring Expression language)

@Autowire

@Qualifier

@Primary

==========

ExcelExportAnnotationConfigS10

code

========

pom.xml

<project xmlns="http://maven.apache.org/POM/4.0.0"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.app</groupId>

<artifactId>ExcelExportAnnotationConfigS10</artifactId>

<version>1.0</version>

<properties>

<maven.compiler.source>21</maven.compiler.source>

<maven.compiler.target>21</maven.compiler.target>

</properties>

<dependencies>

<dependency>

<groupId>org.springframework</groupId>

<artifactId>spring-context</artifactId>

<version>7.0.5</version>

<scope>compile</scope>

</dependency>

</dependencies>

</project>

Spring bean + annotation cofiguration

ExcelExport.java

---

package com.app;

import org.springframework.beans.factory.annotation.Value;

```
import org.springframework.stereotype.Component;
@Component("excel")
public class ExcelExport {
@Value("TEST")
   private String exportName;
@Value(".csv")
   private String fileExt;

@Override
public String toString() {
return "ExcelExport [exportName=" + exportName + ", fileExt=" + fileExt + "]";
}

}
```

MyAppConfig.java

```
package com.app;

import org.springframework.context.annotation.ComponentScan;
import org.springframework.context.annotation.ComponentScans;

@ComponentScan("com.app")
public class MyAppConfig {

}
```

Test.java

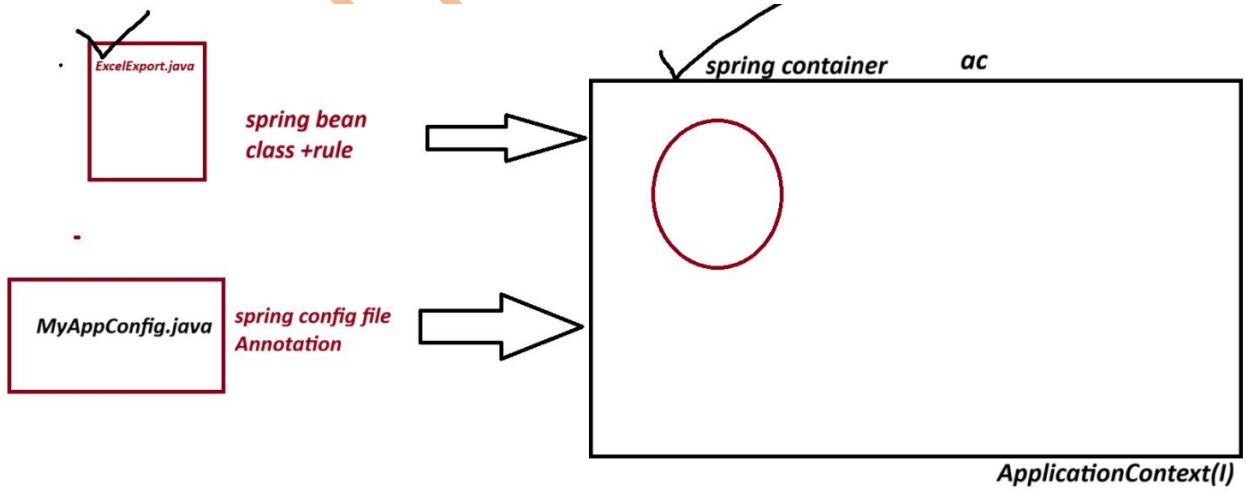
```
package com.app;

import org.springframework.context.annotation.AnnotationConfigApplicationContext;
```

```
import org.springframework.context.ApplicationContext;


public class Test {
public static void main(String[] args) {
AnnotationConfigApplicationContext ac = new AnnotationConfigApplicationContext(MyAppConfig.class);
      // Find classes from a package(basePackage)
//ac.scan("com.app");
      // object creation
     //ac.refresh();



      ExcelExport ee = ac.getBean("excel",ExcelExport.class);
      System.out.println(ee);
}
}
```

output

ExcelExport [exportName=TEST, fileExt=.csv]

Diagram



==========

session 11

========

Spring Annotation Configuration:

1. @Component : create Object to given class

2. @Value : To provide data (Field Injection) to variable

3. @ComponentScan : To find classes using basepakage.

====================Prorperties File=================

_____.properties

=> This is also caleld as key-val pair input as Text format(String)

=> This file is used to provide setup data to application.

=> Here, we can read data from properties file into variable using @Value("${key}")

=>Programmer has to define and load Properties file.

Use @PropertySource annotatio to load file, into spring container.

=> container creates Environment(I) Object (internally used StandardEnvironment(c)).

=> we can read data using @Value("${keyName}")

=> if we define duplicate key with different value (not recomened then last combination is taken into Container)

================

myapp.properties

#Symbol allowed for comment '#'

#key=value

#Symbol allwed to write keyName (dot(.), dash(-), underscore(_))

my.driver=oracle

my.url=oracle-jdbc

my.username=system

my.password=root

my.port=3306

my.active=fALSE

=============

my.port=4444 will be printed

==================

=> For boolean datatype valid input are : false and true

(case-insensitive, False,FASLE,FalSE, True,TRue etc)

========

Project Name: database-connectionS11

Example

code

----

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>database-connectionS11</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

Test.java

```
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.annotation.AnnotationConfigApplicationContext;
```

```
public class Test {

public static void main(String[] args) {
ApplicationContext ac=new AnnotationConfigApplicationContext(MyAppConfig.class);
DatabaseCon obj = ac.getBean("databaseCon",DatabaseCon.class);
System.out.println(obj);
}

}
```

DatabaseCon.java

```
package com.app;

import org.springframework.beans.factory.annotation.Value;
import org.springframework.stereotype.Component;

@Component
public class DatabaseCon {

@Value("${my.db}")
private String driver;

@Value("${my.url}")
private String url;

@Value("${my.username}")
private String username;

@Value("${my.password}")
private String password;
```

```
@Value("${my.port}")
private int port;

@Value("${my.enable}")
private boolean active;

@Override
public String toString() {
return "DatabaseCon [driver=" + driver + ", url=" + url + ", username=" + username + ", password=" + password
+ ", port=" + port + ", active=" + active + "]";
}



}
```

MyAppConfig.java

```
package com.app;

import org.springframework.context.annotation.ComponentScan;
import org.springframework.context.annotation.PropertySource;

@ComponentScan("com.app")
@PropertySource("classpath:myapp.properties")
public class MyAppConfig {


}
```

---

myapp.properties

```
# comment line
# symbol allowed to write keyName (dot . dash - underscore _)
#key=value
my.db=oracle
```

my.url=oracle-jdbc

my.username=indiafreeinternship

my.password=root

my.port=3306

#my.port=3345

#my.port=4444

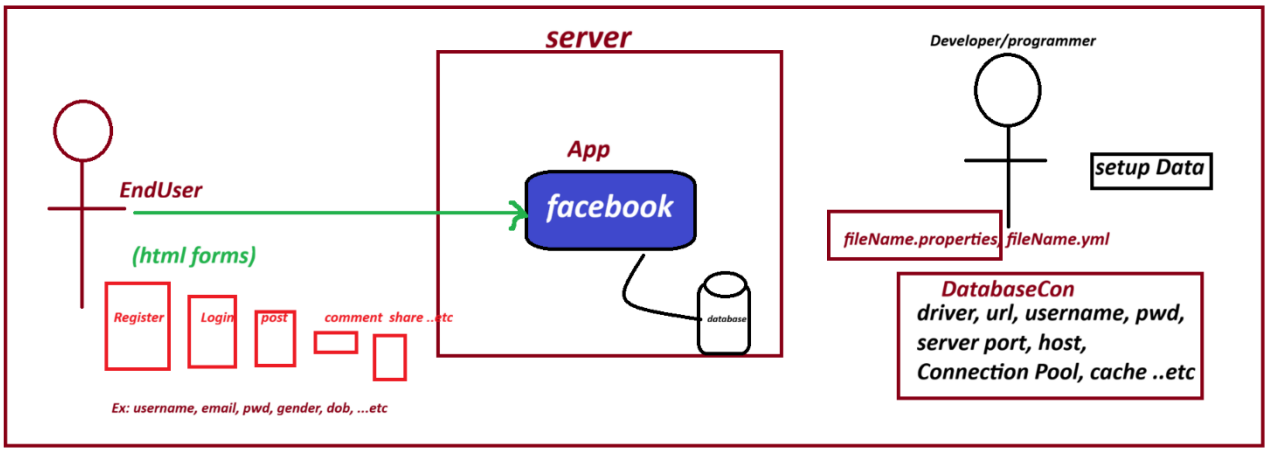
my.enable=tRUe

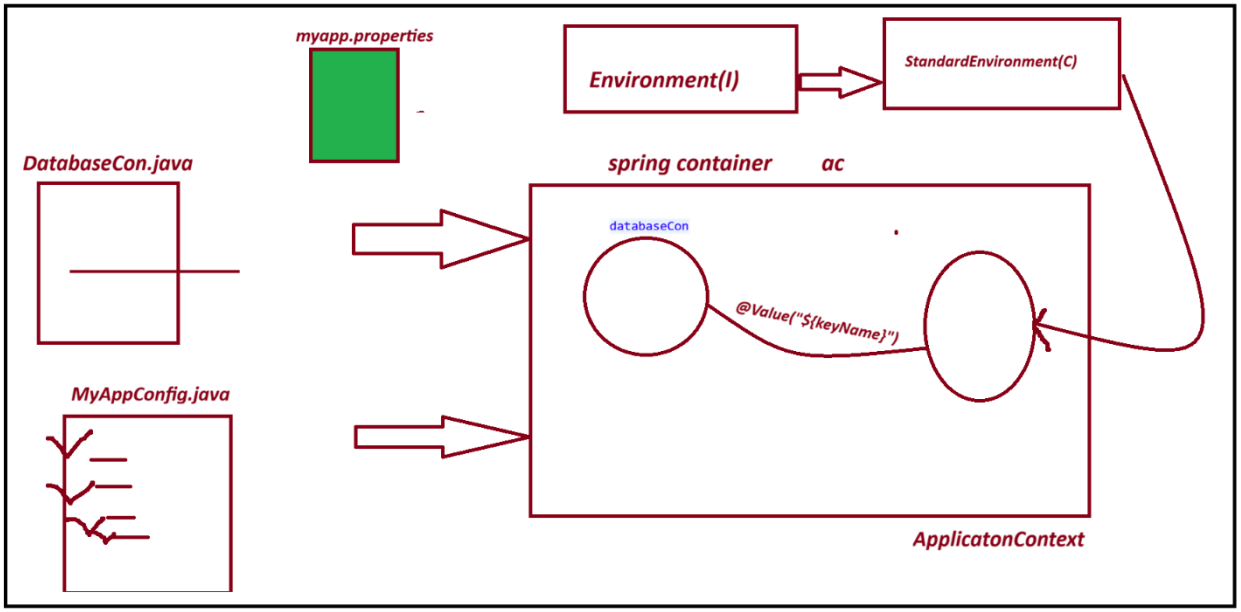
----

output

DatabaseCon [driver=oracle, url=oracle-jdbc, username=indiafreeinternship, password=root, port=4444, active=true]

Diagram





session 12

=======

Spring Annotation configuration

@Component Annotation : To create Object

@Value : Direct value to a variable / Read data from properties

@ComponentScan : back-Package to scan classes

@PropertySource : To Load Properties File

Autowiring

@Autowired Annotation

=> It is a process of Linking objects based on Association Mapping (HAS-A)

=> @Autowired Annotation is used. So, that container can link them.

=>Autowired Annotation Injects Dependecny Beans into Dependent bean.

=> In Case of XML Configuration <ref/> tag is used to link objects.

==========

1. @Autowired will search for child class object.

2. If one child object is found then it is injected.

3. If Zero child object are found then throw exception NoSuchBeanDefinitionException, message like:

At least 1 bean required

==========

---------code----------

project name : AutowiredAnnoS12

==========

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>AutowiredAnnoS12</artifactId>
```

```
<version>1.0</version>
<properties>
<maven.compiler.source>8</maven.compiler.source>
<maven.compiler.target>8</maven.compiler.target>
</properties>

<dependencies>
<!-- Source:
https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>


</project>
```

---

2. Spring bean

EmployeeDao.java

```
package com.app;

import org.springframework.beans.factory.annotation.Value;
import org.springframework.stereotype.Component;

//@Component("edao")
@Component("edao")
public class EmployeeDao {
```

```
@Value("ORACLE")
private String dbName;

@Override
public String toString() {
return "EmployeeDao [dbName=" + dbName + "]";
}

}
```

---

3. spring bean

EmployeeService.java

```
package com.app;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Component;
//ctrl+shift+O
@Component("esobj")
public class EmployeeService {

//@Autowired(required=true)
@Autowired(required=false)
private EmployeeDao dao; // HAS-A

@Override
public String toString() {
return "EmployeeService [dao=" + dao + "]";
}
```

```
}
```

AppConfig.java

```
package com.app;

import org.springframework.context.annotation.ComponentScan;

@ComponentScan("com.app")
public class AppConfig {

}
```

Test.java

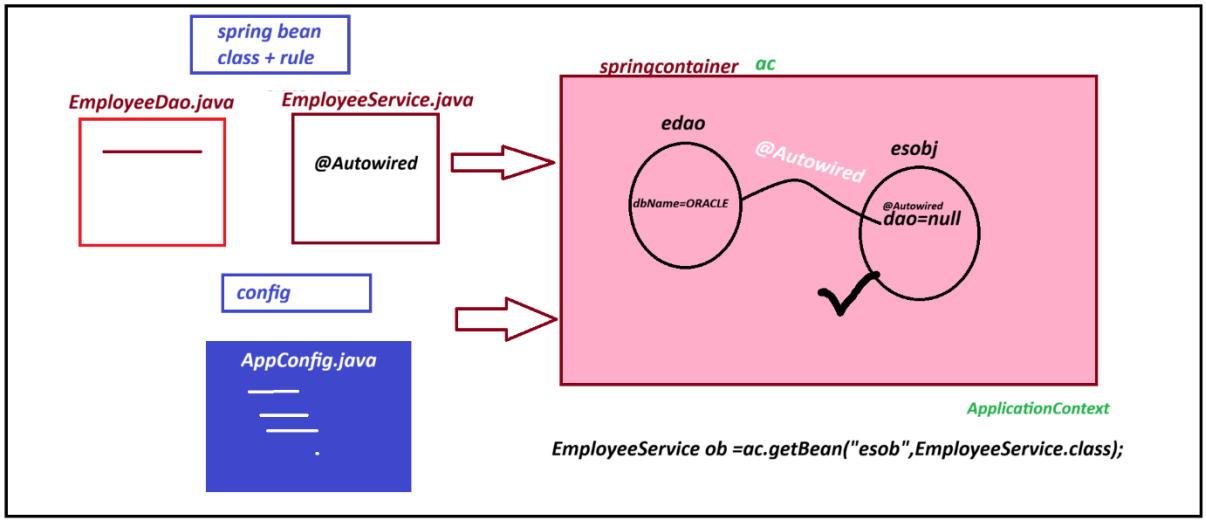
```
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.annotation.AnnotationConfigApplicationContext;

public class Test {

public static void main(String[] args) {
ApplicationContext ac=new AnnotationConfigApplicationContext(AppConfig.class);
EmployeeService ob = ac.getBean("esobj",EmployeeService.class);
System.out.println(ob);

}

}
```

Output

```
EmployeeService [dao=EmployeeDao [dbName=ORACLE]]
```

==========

Diagram



==========

Sessino 13